Homework 2.

1.

$$\text{Dim} = \frac{(8-3) + 2(0)}{1} + 1 = 6 \times 6 \times 4$$

2. $$\text{Dim} = \frac{(10-3) + 2(1)}{1} + 1 = 10 \times 10 \times 2$$

3. $P = (3 \times 3 \times 3 + 1) \times 1 = 28$

4. $$\begin{bmatrix} 2 & 3 \\ 1 & 1 \end{bmatrix}$$

5. $$\begin{bmatrix} 4 & 5 \\ 6 & 7 \end{bmatrix}$$

Part B.

6. CNNs use filters to identified ~~attention~~ patterns in a localized area

7. ensures we capture edges.

8. sharpens images.